

Unless otherwise indicated, each question is of equal value [4 pts]. Clearly show all of your work.

Use $\mathbf{u} = \langle 2, 4, 1 \rangle$ and $\mathbf{v} = \langle 1, -3, 2 \rangle$ to perform the indicated operations in problems 1–3.

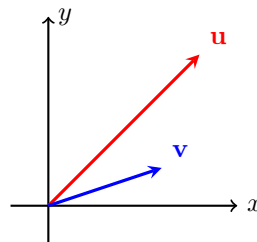
1. $\mathbf{u} - 2\mathbf{v}$

2. $\mathbf{u} \cdot \mathbf{v}$

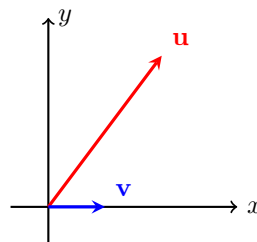
3. $\mathbf{u} \times \mathbf{v}$

Use the graph to draw the following vectors in problems 4–5 (provide labels). Make certain to indicate direction.

4. a. $\frac{1}{2}\mathbf{u}$
 b. $\mathbf{u} + \mathbf{v}$
 c. $\mathbf{u} - \mathbf{v}$ (you do not have to draw it as a position vector)

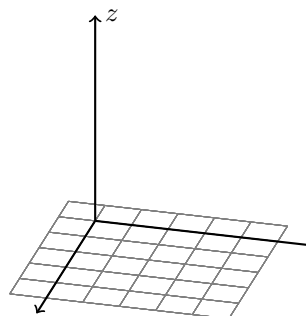


5. a. $\text{proj}_{\mathbf{v}} \mathbf{u}$
 b. $\text{proj}_{\mathbf{u}} \mathbf{v}$



Use the graph to perform the indicated operations (provide labels).

6. a. Use the right-hand rule to label the x and y axes.
 b. Plot the point $(4, 2, 3)$.
 c. Sketch the plane $(x, y, 3)$.



Use the vectors $\mathbf{u} = \langle 3, 6 \rangle$, $\mathbf{v} = \langle 2, 3 \rangle$ and the point $P(1, -4)$ to complete problems 7–10.

7. Find the line through P and parallel to $\mathbf{u}$.
8. Find the angle between $\mathbf{u}$ and $\mathbf{v}$.
9. Find a vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.
10. Is the vector found in problem 9 unique? If not find another vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.

Let $\mathbf{r}(t) = \left\langle 1, \frac{2}{t^2}, e^{-t} \right\rangle$. Evaluate problems 11–14.

11. $\mathbf{r}'(t)$
12. $\int \mathbf{r}(t) dt$
13. $\int_0^1 \mathbf{r}(t) dt$
14. $\lim_{t \rightarrow \infty} \mathbf{r}(t)$

Compute the slope of the tangent line of the vector valued function at $t = 1$.

15. $\mathbf{r}(t) = \langle 1 + t^3, 2 \sin 3t, e^t \rangle$
16. A bullet is fired with a muzzle velocity of 1037 ft/sec from a gun aimed at an angle of 8 above the horizontal. Find a vector describing the vertical and horizontal components of the bullet's velocity.
17. A sailboat floats in a current that flows due west at 3 m/s. The wind is blowing 30 north of east at a speed of 5 m/s. Find the speed of the boat relative to the shore.
18. How much work does it take to slide a box 30 meters along the ground by pulling it with 200 Newtons of force at an angle of 15 from the horizontal?
19. A woman does leg lifts with a 10 lb weight attached to her foot (so the force is directed downward). The distance from her foot to her knee is 2 ft, and her leg below the knee makes an angle of θ to the vertical.
 - a. Find the magnitude of the torque on her knee if she holds her leg at 45.
 - b. What positions minimize and maximize the magnitude of the torque?